A Parkinson's Auxiliary Diagnosis Algorithm Based on a Hyperparameter Optimization Method of Deep Learning

Xingbo Wang, Shujuan Li, Chi-Man Pun, Yijing Guo, Feng Xu, Hao Gao, and Huimin Lu

***Abstract*—Parkinson's disease is a common mental disease in the world, especially in the middle-aged and elderly groups. Today, clinical diagnosis is the main diagnostic method of Parkinson's disease, but the diagnosis results are not ideal, especially in the early stage of the disease. In this paper, a Parkinson's auxiliary diagnosis algorithm based on a hyperparameter optimization method of deep learning is proposed for the Parkinson's diagnosis. The diagnosis system uses ResNet50 to achieve feature extraction and Parkinson's classification, mainly including speech signal processing part, algorithm improvement part based on Artificial Bee Colony algorithm (ABC) and optimizing the hyperparameters of ResNet50 part. The improved algorithm is called Gbest Dimension Artificial Bee Colony algorithm (GDABC), proposing "Range pruning strategy" which aims at narrowing the scope of search and "Dimension adjustment strategy" which is to adjust gbest dimension by dimension. The accuracy of the diagnosis system in the verification set of Mobile Device Voice Recordings at King's College London (MDVR-CKL) dataset can reach more than 96%. Compared with current Parkinson's sound diagnosis methods and other optimization algorithms, our auxiliary diagnosis system shows better classification performance on the dataset within limited time and resources.**

***Index Terms*—Artificial bee colony algorithm, deep learning, hyperparameter optimization, parkinson's auxiliary speech diagnosis.**

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I. Introduction

Parkinson's disease is considered to be the most serious neurodegenerative disease besides Alzheimer's disease [1]. Parkinson's disease (PD) mainly manifests in tremor, movement and sound retardation, such as language disorders and handwriting change [2]. PD is difficult to detect in the early stage and there is no good treatment method at present [3]. Although medical devices can reflect EEG signals, these signal changes may be caused by other reasons instead of PD. Another fact is that medical devices specifically targeting PD are not common in hospitals now. The final diagnosis is mainly be based on the professional knowledge of the doctor. However, the response of patients to dopaminergic therapy is not clear and the signs of atypical PD may not appear, which are not conducive to the doctor's diagnosis and lead to the low accuracy of clinical diagnosis in the early stage. The accuracy of clinical diagnosis is only 26% at the first visit [4]. Therefore, it is urgent to find an effective auxiliary diagnosis technology.

Recently, researchers found that most patients with PD appear language disorders at an early stage, such as dysphonia, dysarthria [5] and these symptoms will worsen over time. These sound symptoms are easier to detect in time than changes in handwriting. Today, voice acquisition technology and sound signal processing technology are becoming more and more mature, allowing to collect speech and process sound signals for patients with PD. Therefore, many scholars have begun to turn their attention to the auxiliary diagnosis of Parkinson's sound.

The auxiliary diagnosis of PD is roughly divided into three steps: data preprocessing, feature extraction, and classification. The most commonly used method for Parkinson's diagnosis is to extract features using traditional feature extraction methods and implement classification with a machine learning classifier. Mel Frequency Cepstral Coefficients is currently the most mainstream feature extraction method, but it has disadvantages such as complex parameters, poor feature representation of high-order audio, and the presence of missing features or feature redundancy [6]. The traditional machine learning classifiers also have some problems, such as high data requirements, difficult parameter adjustment, large amount of calculation and model instability [7]. In recent years, deep learning has become a powerful tool, especially deep neural network is widely used in various fields such as image, speech, medical, natural language processing [8]. Deep learning learns higher-level and high-dimensional features through training data without human intervention [9], [10]. In addition, deep neural network has also been proved to be a very effective classifier for PD diagnosis [11]. Using deep learning methods to classify PD has become a feasible and effective direction of Parkinson's diagnosis. Therefore, instead of traditional scheme, we directly implement feature extraction and classification using deep neural network in our auxiliary diagnostic system.

In recent years, disease diagnosis based on deep learning has gradually attracted scholars' attention [12], [13] due to the excellent performance of deep learning in different fields. In [14], information gain is used for feature selection and artificial neural network is used to classify the diagnosis of patients. The verification accuracy of UCI Parkinson sound dataset [15] is 83.3%. In [16], five simple artificial neural networks were used to classify PD on UCI dataset, and Pearson and Kendall's correlation coefficients were used for feature selection, with the highest accuracy of 67.25% and 86.47% respectively. [17] used stacked automatic encoders to classify PD on UCI dataset and found that deep neural network has better classification accuracy than other classifiers such as support vector machine and decision tree. [11] used grid search to adjust the hyperparameters of the neural network model. It was found that the accuracy of the optimized model optimized on UCI dataset is as high as 91.69%. These researches demonstrate that deep learning performs well for Parkinson's diagnosis, but it is still difficult to select the parameters of neural network model. There are some efforts to improve the accuracy of Parkinson's diagnosis by optimizing the hyperparameters of the model, but the effect is not ideal. Currently, many studies employ evolutionary computing (EC) methods to automatically improve and optimize DL [18], [19], including optimizing network hyperparameters. ABC has become a hot research topic for scholars due to its simple parameter setting with powerful global search ability [20]. In related works [21], [22], ABC also has demonstrated good performance in optimizing the hyperparameters of neural network. Based on this, we employ ABC to optimize the hyperparameters of our deep learning based diagnostic system. Parkinson's diagnosis is actually a binary classification problem. However, due to the complex and difficult extraction of Parkinson's sound features, the CNN in this auxiliary diagnosis system requires a deep network. ResNet50 can not only extract the features of the data well, but also its residual structure can solve the problem of network degradation [23]. And it is also commonly used in medical diagnosis [24]. Therefore, we choose ResNet50 as the network model.

In our work, we first pre-processe the sound dataset and then propose GDABC to optimize hyperparameters of ResNet50. Finally, the optimized ResNet50 is used or feature extraction and classification on the MDVR-KCL Parkinson sound dataset. The major contributions of this paper include:

1) A Parkinson's diagnosis system which uses GDABC to optimize ResNet50 hyperparameters is proposed. This is the first time using ABC to optimize the hyperparameters of deep neural network for Parkinson sound classification.
2) We propose GDABC, which mainly adds "Mixed coding strategy," "Range pruning strategy" and "Dimension adjustment strategy " on the basis of basic ABC. "Mixed coding strategy" maps different types of hyperparameters to continuous domain. "Range pruning strategy" mainly uses gbest to narrow the scope of hyperparametric search. "Dimension adjustment strategy" is to adjust gbest dimension by dimension. For integer and floating-point hyperparameters, the strategy adopts different dimension update methods.
3) We explored the performance of convolutional neural network on the MDVR-KCL dataset and compared the performance with other swarm intelligence optimization algorithms. In addition, we also compare our work with current methods in Parkinson sound detection. Experiments demonstrate that our method performs best in limited time and resources.
4) Our system does not use other commonly used voice feature extraction techniques, but utilizes the feature extraction and classification capabilities of deep network to extract the features of the MDVR-KCL dataset and achieve Parkinson's classification.

The arrangement of this paper is as follows: some related work, ResNet50 and the basic ABC are introduced in Section II. Next, proposed GDABC are introduced in detail in Section III. Then, we conduct experiments and compare the experimental results with the current works in Section IV. Finally, we draw a conclusion in in Section V.

II. Related Work

A. DL Based Parkinson Auxiliary Diagnosis

In recent years, deep learning has developed rapidly and has been widely used in various clinical applications [25], [26]. PD is difficult to diagnose and cure, researchers have also begun to apply deep learning methods to the diagnosis of PD. Li et al. [27] proposed an auxiliary diagnosis scheme based on continuous convolutional network (CC-Net) for automatic classification of Parkinson's in Archimedes' template-free spiral hand-painting. This method better helped PD patients to complete the early diagnosis and its classification accuracy was about 89.3%. In [28], Afonso et al. modeled the handwriting dynamic signal as an image through the recursive graph method and used CNN to achieve Parkinson's Diagnosis. In another work [29], a 3D convolutional diagnostic PD scheme was proposed to perform Parkinson's classification diagnosis on preprocessed MRI image data. The results proved that compared with the traditional feature extraction technology, the features obtained by the deep learning model were more conducive to the diagnosis of PD. In [30], GAN was used to enhance MR image data, and the pretrained AlexNet model was used for predictive diagnosis. The fine-tuned network had an average classification accuracy of 89.23%. Berus et al. [16] used the Pearson and Kendall correlation coefficients to perform feature selection on Parkinson's speech dataset and used a feedforward artificial neural network (ANN) to predict PD of the tested individuals, with a final test accuracy of 86.47%. In [31], both traditional decision trees and residual neural networks were used for the classification and diagnosis of Parkinson's voice dataset, and the results proved that residual neural networks can effectively distinguish PD patients from healthy people. Another work [32] was to diagnose Parkinson's sound dataset using the ResNet architecture which was pretrained on ImageNet and SVD databases. The diagnostic accuracy obtained on the validation set was higher than 90%. Fatihs and other researchers [33] constructed a LSTM network model for PD diagnosis based on speech impairment, applied end-to-end training of the network without using any feature selection method, and obtained a prediction accuracy of 94.27%.

B. Hyperparameter Optimization of DL Based on EAs

Deep learning has attracted extensive attention of scholars due to its powerful feature extraction and classification capabilities in image and speech and et al. [34]. In recent years, COVID-19 has swept the world, and the use of deep learning in medical diagnosis [35], [36] has also been pushed to a climax. The performance of DL is greatly dependent on the hyperparameter settings of the network, and manual parameter adjustment is time-consuming and labor-intensive. Evolutionary algorithms which can effectively optimize complex optimization problems, are gradually applied to hyperparameter optimization of deep learning [37]. In a recent study [38], a randomly occurring distributed delayed PSO (RODDPSO) algorithm was proposed to optimize the hyperparameters of the DBN. And the optimized network successfully classified patient visit dispositions in the accident & emergency department of a hospital. Another study [39] proposed a genetic algorithm to optimize 4 hyperparameters of the Residual Network which was to diagnose diabetic retinopathy. The final diagnostic accuracy was 93.8%. In [40], Mohakud and his collaborators proposed a novel EN-GWO algorithm based on a neighborhood search strategy to optimize a fully convolutional encoder-decoder network for the segmentation of skin cancer images. Experiments demonstrated that the optimized model outperformed other state-of-the-art deep learning models. In an excellent work [21], Karthiga et al. combined the exploration ability of ABC and the development ability of grey wolf optimizer (GWO) to propose a hybrid GWO-ABC. The algorithm was used to optimize 7 hyperparameters of the convolutional neural network. The optimized model realized the prediction of cardiovascular disease, and successfully improved the detection accuracy by 6.16%. In a work of liver cancer auxiliary diagnosis [41], a new bionic deep learning method was proposed to optimize the prediction results of liver cancer. This work mixed SegNet and UNet, and used ABC to adjust the hyperparameters of the hybrid network. Guo et al. [22] proposed a diagnosis system driven by the deep CNN model and adjusted the hyperparameters of model using ABC. The diagnostic system performed two classification on CXR images used in the diagnosis of lung abnormalities and showed satisfactory results.

C. ResNet50

In recent years, CNN has also been widely used in sound processing and enhancement, as well as in disease detection [31].

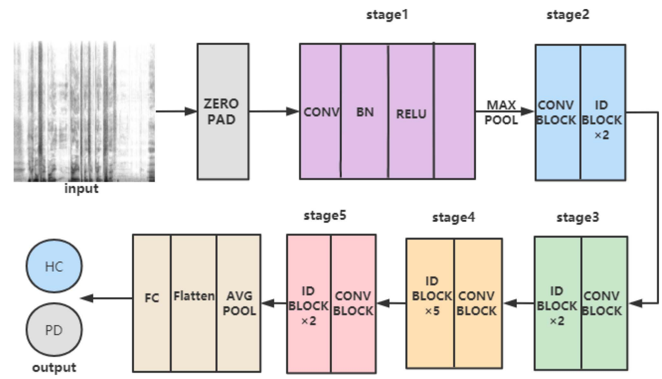

Fig. 1. ResNet50 network structure diagram: The network includes a total of 5 stages. Its input is the spectrogram generated after preprocessing and its output is a label of 0, or 1, 0 means healthy, 1 means Parkinson.

Especially the residual structure of ResNet50 can solve the problem of network degradation, becoming a common deep network in medical diagnosis. ResNet was proposed by Kaiming He in 2015 [42]. The key of ResNet is putting forward the idea of Residual block. H (x) represents the network output y, x represents the shortcut input, ResNet learns not the ordinary feature output y, but the residual function F (x) = H (x) - x [43]. Scholars have found that the deep network can better learn the characteristics of data, but with the deepening of the network, there will be the phenomenon of network degradation. So that the network can not continue to learn. Therefore, ResNet proposes to overlay the map y = x on the shallow layer, which can increase the network depth without causing network degradation and reduce computational complexity [23]. In ResNet50, there are two types of residual blocks: identity block and conv block. Identity block's shortcut is an identity mapping and its output channels equal to input channels, conv block's shorcut consists of a convolution layer and a BN layer to make dimensional changes. The network structure of ResNet50 is shown in Fig. 1.

Although ResNet can deepen the number of network layers without disappearing the gradient, it can not be guaranteed to avoid the reduction of model accuracy which is largely determined by the hyperparameters of the network. However, the hyperparameter setting required by the same network model for different datasets is also different. That is to say, using the standard ResNet50 with the default hyperparameter setting may not achieve the best effect of the model on the MDVR-KCL dataset. Therefore, we optimize the hyperparameters of the standard ResNet50 and find a set of hyperparameters that are most suitable for the dataset to achieve the ideal classification accuracy.

D. The Basic ABC

In 2005, ABC was first proposed by Karaboga [44] inspired by the behavior of bee colony collecting nectar. Artificial bee colonies are divided into three categories: Employed bees, Onlooker bees and Scout bees. The number of employed bees is the same as that of onlooker bees, which is equal to the number of nectar. The task of Employed bees is to search and mine the nectar. Onlooker bees are to mine the nectar locally. And Scout bees are responsible for exploring the new nectar. Bees exchange the information of nectar by waggle dance in dancing area. Each bee abandons the previous nectar and moves to its current optimal nectar according to the greedy strategy. The specific steps of basic ABC are as follows.

1) Initialization phase

Randomly generate SN nectar, which is also equal to the number of employed bees. The fitness function value of each nectar is calculated by Formula 1.

$$x_{i,j} = x_j^{\min} + rand(0,1) \cdot \left(x_j^{\max} - x_j^{\min}\right) \quad (1)$$

$2 * SN$ is the population number, $j \in D$ is the dimension, $i \in \{1, 2, \ldots, SN\}$. $x_j^{\min}$, $x_j^{\max}$ is the j-th dimension of the lower bound and upper bound of the nectar location, respectively.

2) Employed bee phase

After the initialization phase, the employed bees will find the nectar in the whole nectar space through the update strategy 2 and remember the current information of nectar so as to communicate with other bees in the future.

$$v_{ij}^{t+1} = x_{ij}^t + \psi \cdot \left(x_{ij}^t - x_{kj}^t\right)) \quad (2)$$

Formula 2 represents updating the j-th dimension of the i-th feasible solution, and v_{ij}^{t+1} represents the j-th dimension of the updated nectar. Where, $j \in D$ is the dimension, $k \in \{1, 2, \ldots, SN\}$ is a randomly selected index and must be different from i; ψ is a random number with a value range of $[-1, 1]$.

3) Onlooker bee phase

In onlooker bee phase, according to the roulette mechanism, the onlooker bees choose to an employed bee to follow and look for new nectar locally near the shared nectar location information by the selected employed bee through Formula 2. The number of onlooker bees is also SN. Before that, the probability value of each nectar is calculated according to Formula 3. Roulette mechanism means that the greater the fitness value of nectar, the greater the probability of being selected.

$$P_i = \frac{fit_i}{\sum_{j=1}^{SN} fit_j} \quad (3)$$

fit_i represents the fitness value of the i-th nectar and P_i represents the probability that the nectar is selected by the onlooker bees.

After the onlooker bees develop the nectar locally, the greedy strategy is used to update the nectar. In other words, if the fitness value of the found nectar is better than that of the previous nectar, the previous nectar will be abandoned and replaced by the new one. Otherwise the found nectar will be abandoned and the original position will be retained.

4) Scout bee phase

In the later stage of nectar collection, the whole nectar search may fall into the local optimal value. After the max trial is exceeded, the scout bee abandons the current nectar and searches for new nectar within the whole nectar range according to Formula 1. Trials Limit is the max trials parameter which weighs the global search and local development ability of the algorithm. It is also the only control parameter to be set except the population number.

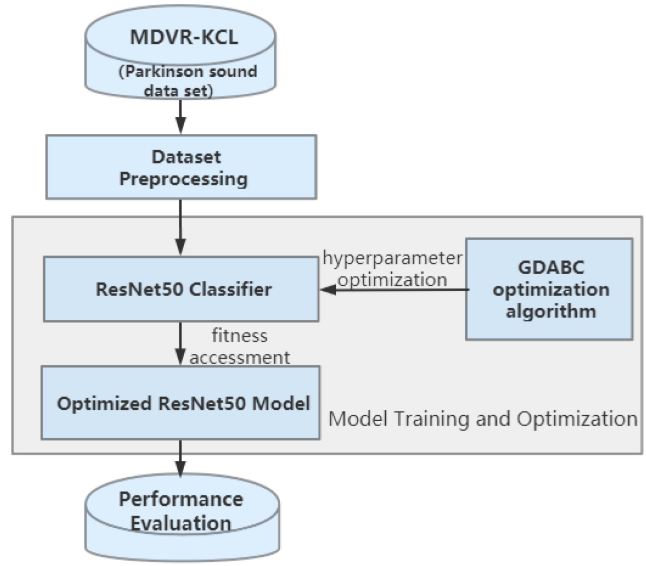

Fig. 2. Overall framework of Parkinson's diagnostic system. The basic network framework is ResNet50, and the ABC is improved. The improved algorithm is used to optimize the network hyperparameters, and the optimal network model is used to achieve Parkinson's classification.

III. Parkinson's Auxiliary Diagnosis System

Fig. 2 illustrates the framework of our proposed automatic diagnosis system of PD. The MDVR-KCL Parkinson's sound dataset is first converted into spectrograms after preprocessing. Then the GDABC based on ABC is used to search the appropriate hyperparameter setting for ResNet50, and the fitness is evaluated on the validation set. The best hyperparameter setting is selected according to the fitness which represents the classification accuracy. Finally, the optimized ResNet50 model is evaluated on the test set to verify the effectiveness of the system. In the whole auxiliary diagnosis system, the optimized ResNet50 is responsible for sound feature extraction and Parkinson's classification. The specific algorithm flow is shown in Algorithm 1. In this part, data preprocessing and proposed GDABC optimization algorithm are introduced in detail.

A. Pre-Processing

In order to eliminate the interference of ambient background noise and present the vocal information [45] of volunteers, we preprocess the sound dataset to generate the spectrogram from the original sound, which will be more conducive to the training of ResNet50 later. To make the spectrogram retain more formant information, before preprocessing, we cut each volunteer's audio into 5-second segments. After cropping, audio segments are generated. The general flow of pre-processing for these audio segments is illustrated in Fig. 3. The energy of high-frequency signal is usually low, so it is necessary to increase the energy

Algorithm 1: Overall flow of Parkinson's Auxiliary Diagnosis System.

Input: the MDVR-KCL dataset

1 After pre-processing, the audio data is converted into spectrograms;
2 Hyperparameter optimization of ResNet50;
3 The best hyperparameter setting is selected according to the fitness value;
4 The optimized ResNet50 obtained by GDABC is evaluated on the test set again;

Output: optimized ResNet50 model in MDVR-KCL Parkinson's sound data set

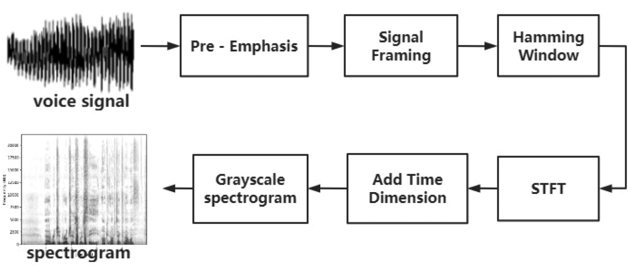

Fig. 3. The general flow of pre-processing. The input is voice wav, the output is grayscale spectrogram.

of high-frequency part through pre-emphasis. Next, the sound signal is divided into frames. Usually, the signal is divided into 20-40 ms frames. Then, the window function is used to smooth the signal and short-time Fourier transform (STFT) is performed on each frame signal, converting per frame from time domain to frequency domain. Finally, the frequency domain information of each frame is stitched together in time to obtain the sound spectrum map. In the end, grayscale spectrograms are obtained. The spectrogram includes three dimensions: time, frequency and amplitude. The x-axis of the spectrogram is time, the y-axis is frequency, and the color is the magnitude of the frequency. Because the resolution of the standard input image of ResNet50 is $256 \times 256 \times 3$, this paper sets all spectrograms to the same resolution to maximize the performance of the network.

B. Proposed GDABC Optimization Algorithm

In recent years, as one of the typical intelligent optimization algorithms, ABC has been applied to solve a variety of problems because of its simple parameter setting and good global search ability. However, its local search ability is weak and its convergence is relatively slow [46]. So we choose ABC to optimize the hyperparameters of ResNet50, and improve the basic ABC based on gbest. Gbest in this paper represents the best individual the algorithm ever achieved, that is, the best set of hyperparameters found by the algorithm.

In this part, in order to improve the shortcomings of the basic ABC, the GDABC is proposed. Taking advantage of powerful exploring ability of the basic ABC algorithm, the GDABC algorithm innovatively proposes the "Range pruning strategy" and the "Dimension adjustment strategy," which not only improve exploiting and convergence ability of the ABC algorithm but also improve the accuracy of the optimized model in Parkinson's classification. The overall algorithm flow of GDABC is illustrated in Algorithm 2. First, the mixed variable coding scheme is introduced due to the different types of hyperparameters of optimized ResNet50. Second, the "Range pruning strategy" is introduced in detail to obtain a pruned new search range. Third, the "Dimension adjustment strategy" is proposed to update the position of gbest for finding a better solution.

1) Mixed Variable Coding Strategy: In our work, we optimized 8 hyperparameters of ResNet50, including training hyparameters and model hyparameters. The detailed optimized hyperparameters are shown in Table I. In other words, the variable is an 8-dimensional array and each dimension represents a hyperparameter in this paper. Among these hyperparameters, filter size, optimizer, activation function and pooling type are discrete values, while the other hyperparameters are continuous

Algorithm 2: Overall Flow of GDABC.

```
Input: population size, maximum generation,
       dimension,Trails Limit,Stagnation Limit,
       Upper and Lower limits
Use the Mixed Coding strategy to encode the 8
  hyperparameters of ResNet50 ;
Carry out random initialization of nectar according
  Formula (1), and evaluate the fitness of nectar;
for generation=1 to generation =10 do
    if gbest is stagnation then
        while stop condition is false do
            Calculate the new upper and lower limits
              of new nectar according to the
              Range Pruning strategy;
            for dim=1 to dim=8 do
                Produce new nectar and fine tune
                  gbest according to Formula (4) of the
                  Dimension Adjustment strategy;
                if new gbest acc > gbest acc then
                    Update gbest according to greedy
                      algorithm;
                end
                else
                    Set stop condition is True
                end
            end
        end
        break
    end
    else
        Employed bee phase();
        Onlooker bee phase();
        Scout bee phase();
    end
end
Output: relatively optimal parameter setting for
        ResNet50 and fitness evaluation number
```

TABLE I
THE HYPERPARAMETERS OF RESNET50 TO BE OPTIMIZED

Variables	Hyperparameter	Range
x_0	the filter size of the first convolution layer	3, 4, 5, 6, 7
x_1	optimizer	SGD, Adam, RMSprop
x_2	activation function	relu, sigmoid, tanh
x_3	pooling type	AveragePool, MaxPool
x_4	weight decay rate	[0.0001, 0.01]
x_5	momentum	[0.9, 0.95]
x_6	initial learning rate	[0.001, 0.1]
x_7	dropout rate	[0.3, 0.5]

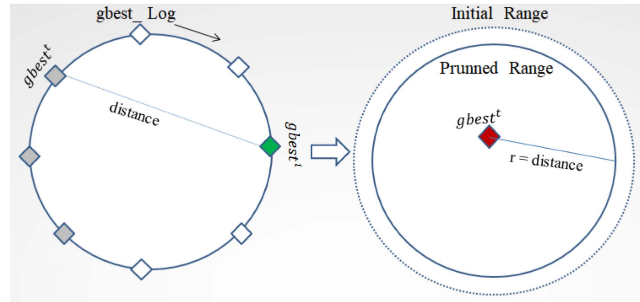

Fig. 4. Sketch map of Range pruning strategy. Use $gbest^t$ and $gbest^i$ to obtain the $Gbest\ Distance$, and then use the $gbest^t$ as the base value to expand the $Gbest\ Distance$ around to obtain the $Pruned\ Range$.

numbers. The ABC algorithm is mainly suitable for continuous domain optimization, but not for discrete domain optimization. Therefore, in order to better utilize the advantages of the algorithm, we proposed the mixed variable coding strategy to map discrete hyperparameters to continuous space.

First, the hyperparameters of discrete type are encoded into integers. For example, three different optimizer types are respectively encoded as 0, 1, 2. Then, the hyperparameter of discrete type is viewd as a variable of a continuous domain. For example, the filter size can be any value in the range of (3, 7), such as 3.1. In other words, the continuous domains of these four discrete hyperparameters respectively are [3.0, 7.0], [0.0, 2.0], [0.0, 2.0] and [0.0, 1.0] at this moment. The continuous domains of the other four continuous hyperparameters are shown in Table I. And these eight hyperparameters can be coded to any value in their continuous domain, such as [3.2, 1.1, 0.6, 0.7, 0.0005, 0.9, 0.01, 0.35]. In this way, the 8 hyperparameters are mapped to the continuous field. However, we round the values of the discrete hyperparameters to meet the parameter requirements of neural network when training model. In other words, the discrete hyperparameter which is converted into continuous domain is restored to discrete domain when network with this set of hyperparameters is trained. Its discrete domain is shown in Table I. That is, the set of hyperparameters changes back to [3, Adam, sigmoid, MaxPool, 0.0005, 0.9, 0.01, 0.35] when network is trained. After adopting the mixed variable coding strategy, hyperparameter optimization becomes an optimization problem in the continuous domain, which fully conforms to the original design of ABC. So its corresponding formula can be directly applied to update the hyperparameters. This simplifies the process of optimization without designing specialized formulations for hyperparameter update of discrete type.

2) Range Pruning Strategy: The range pruning strategy improves the convergence ability of the algorithm on the one hand, and paves the way for the dimension adjustment strategy on the other hand. The $gbest\_Log$ which is a memory matrix records the information of each generation of gbest in the search process. When the algorithm converges, that is, the best nectar has not changed for several consecutive generations which is defined this as gbest stagnation. The gbest stagnation means that the ABC algorithm is trapped into a local optimum, and then its precision should be enhanced. The search space of each onlooker bee in the basic ABC is between its following employed bee and another randomly selected employed bee which enhances the local search ability of the algorithm but sacrifices the convergence of the algorithm. Therefore, our onlooker bees choose to follow only an optimal nectar, and we also redesign the search space based on $gbest\_Log$, which not only accelerates the convergence of the algorithm, but also retains the local search ability of the algorithm. The search space redesigned according to $gbest\_Log$ is called range pruning strategy.

In $gbest\_Log$, we look for a different gbest which is the one closest to the stagnant best and different from the stagnant best in $gbest\_Log$. The different gbest is called $gbest^i$, which is marked in green in the figure. The stagnant gbests are marked in grey in the figure. The difference between the stagnant best $gbest^t$ and $gbest^i$ is defined as $Gbest\ Distance$, which is the absolute value of the difference in the corresponding dimensions of the two gbests. The dimension of $Gbest\ Distance$ is 8, equal to the dimension of the variable. The range of the circle with the stagnant gbest value as the center and $Gbest\ Distance$ as the radius is the new search range. For continuous hyperparameters, the the new search range is called $Pruned\ Range$. To further enhance the exploiting ability for discrete hyperparameters, we add two units of floating space on the basis of its new search range. In other words, for discrete hyperparameters, $Pruned\ Range$ is the range after being added. Its formula is shown in Formula 4 and l represents a unit length. Of course, $Pruned\ Range$ cannot exceed the initial upper and lower bound of the algorithm. The schematic diagram of this strategy is illustrated in Fig. 4. In the figure, i represents the ith generation. If gbest has not changed for three consecutive generations in this paper, we think it has stagnated. In other words, Stagnation Limit parameter of GDABC is 3.

$$Pruned\ Range = Gbest\ Distance + [-2\,l, +2\,l] \tag{4}$$

3) Dimension Adjustment Strategy: After the range pruning strategy, we get $Pruned\ Range$. When the optimal value stagnates, considering the convergence, our onlooker bees choose to follow only the optimal nectar and fine tune the optimal nectar dimension by dimension according to $Pruned\ Range$. This will improve not only the exploiting ability of the whole algorithm but also the performance of the optimized model.

Algorithm 3: Integer Strategy of the Dimension Adjustment Strategy.

Input: the bee_Log memory matrix, the $Pruned\ Range$ and gbest

for *i in [0,7]* **do**
  for v *in* $Pruned\ Range$ **do**
    change the ith dimension of the current gbest to v, constituting the positon of new nectar $nectar_v$;
    if $nectar_v$ *in* bee_Log **then**
      call the fitness of the same solution $nectar_v$ in the memory matrix ;
    end
    else
      train the network model to obtain the fitness of $nectar_v$;
      add the location of the new nectar and its corresponding fitness into the bee_Log ;
    end
  end
end

Output: :the position and fitness of new nectars

We adopt two different adjustment strategies in the dimension adjustment part. For hyperparameters of discrete type, we adopt Integer Strategy to evaluate each value within its search range. At the same time, keep the hyperparameter values of the other dimensions unchanged. However, for hyperparametesr of continuous type, we adopt Float Strategy to increase the chance of jumping out of the local optimum. The specific dimension adjustment strategies are as follows:

Integer Strategy: We use bee_Log to record the position of each nectar and the corresponding fitness information at each iteration. When the hyperparameter type is discrete, each value will be iterated over in $Pruned\ Range$. The formula is shown as (5). $Pruned\ Range$ can be calculated by the Formula 4 and v is any value in the range. The other hyperparameters of the current gbest keep unchanged, and the discrete hyperparameter becomes a new value, constituting the location of new nectar. This will search for the best value within its search range as much as possible. Then before evaluating the fitness of the new nectar, traverse bee_Log memory matrix. If the location of new nectar is found in bee_Log, directly call the fitness of the same solution in the matrix, which saves the time of network training and accelerates the convergence of the algorithm. Otherwise, the fitness of the new nectar will be evaluated by training the network. Finally, add the location of the new nectar and its corresponding fitness to bee_Log. The process of Integer Strategy is illustrated in Algorithm 3.

$$gbset_j^{t+1} = gbset_j^t + v, v \in Pruned\ Range \tag{5}$$

Float Strategy: Gaussian randomization is used to fine tuning of gbest in $Pruned\ Range$ when the hyperparameter type is continuous, which avoids falling into the local optimal value. The specific gbest search scheme is described as Formula 6. The distance is $Gbest\ Distance$ obtained in range pruning strategy. We use the current gbest and distance to generate new values around the current gbest value, which can not only retain the best information of current population but also use distance to jump out of local optimal value.

$$gbest_j^{t+1} = Gauss\left(gbest_j^t, distance\right) \tag{6}$$

IV. Experiment and Discussion

In this part, first introduce the Parkinson's sound data set and experiment setup. Next, the experiment setup is introduced in detail. And then we conduct experiments in two directions. Finally, we discuss and analyze the experimental results.

A. Dataset

Most of the sound datasets used for Parkinson's diagnosis are vowel pronunciation test or speech test for volunteers, such as common UCI Parkinson sound dataset. Due to the complexity of the MDVR-KCL dataset collection, there are few studies on the sound data set for Parkinson's diagnosis. And most of the current methods for the diagnosis of Parkinson disease also perform poorly on this dataset. Therefore, we choose to improve the diagnostic accuracy on this dataset to prove the effectiveness of our work. The MDVR-KCL dataset was recorded at King's College London (KCL) hospital in 2017. The dataset is the voice sample obtained from 21 healthy controls (HC) and 16 PD patients through smart phones. The speech test is different from general Parkinson speech dataset that only reads vowels to the subjects. It includes two parts: first let the subjects read the poem, north wind and sun, and then have a spontaneous dialogue with the subjects. The original sound dataset is close to 1.27 GB and is saved in wave file format (. WAV). The sound data is marked according to the following scheme: SiHS-HYR-UPDRS-II-5-UPDRS iii-18. We use HS health status label (corresponding HC or PD) as the label of data and define 0 as healthy and 1 as Parkinson's patient. After preprocessing, a total of 2026 grayscale spectrograms were available, including 1187 from healthy subjects and 839 from Parkinson's patients. The specific preprocessing process is in Section III-A.

B. Experimental Setup

We use the proposed GDABC algorithm to adjust 8 hyperparameters of ResNet50 on the MDVR-KCL dataset. After that, obtain the fitness evaluation number and take it as the maximum evaluation times of other optimization algorithms. As a comparison, under the same conditions and objectives, we first use other swarm intelligence algorithms such as PSO [47], QPSO [48], DE [49] and the basic ABC algorithm [44] to tune ResNet50 on the MDVR-KCL dataset. In addition, we compare our algorithm with other work on the MDVR-KCL sound dataset. All methods compared with our work have the same network parameter settings. This section mainly introduces our network training setting and related intelligent optimization algorithm settings.

In our work, the MDVR-KC dataset is divided into training set, verification set and test set according to the ratio of 8:1:1. Our objective evaluation function is the classification accuracy of adjusted ResNet50 on the verification set. For each fitness

TABLE II
THE GDABC ALGORITHM AND NETWORK PARAMETER SETTINGS

Parameter	Value
GDABC	
population size	10
maximum generation	10
Trails Limit	3
Stagnation Limit	3
Network Training	
epoch	50
batchsize	32

TABLE III
THE PAREMETER SETTING OF PSO,QPSO,DE,ABC ALGORITHM

Algorithm	Parameter Setting
PSO	$\omega = 0.9 \sim 0.4$, $c_p = c_g = 1.5$
QPSO	$\beta = 0.6$
DE	$F = 0.5$, $CR = 0.8$
ABC	$Trails\ Limit\ = 3$

evaluation, train ResNet50 for 50 epochs with a batch size of 32. In the process of network training, in order to make the model converge better and faster, we use the learning rate reduction strategy. When the accuracy on the verification set does not change within 5 consecutive epochs, reduce the learning rate to half. Both the population size and maximum generation are set as 10. And Trails Limit of ABC is 3. The GDABC algorithm and network parameter settings of our work are shown in the Table II. The parameters of compared swarm intelligence algorithm are set according to experience, shown as Table III.

Our codes which use keras library are run on NVIDIA Tesla with 24 G video memory and the framework is tensorflow.

C. Experimental Results and Disscussion

The experiments are divided into two categories. The first experiment replaces the GDABC algorithm in our Parkinson's auxiliary diagnosis system with other typical algorithms to compare the results on the MDVR-KCL dataset, which mainly verifies the effectiveness of the proposed GDABC. The second experiment compares other Parkinson's diagnosis works on the MDVR-KCL dataset with our work to prove the effectiveness and progressiveness of our entire auxiliary diagnosis system. The following are the results and analysis of experiments.

Experiment 1. We compare the Parkinson classification results of GDABC, ABC [44], PSO [47], QPSO [48] and DE [49] algorithms on the MDVR-KCL Parkinson sound dataset. The iteration times of GDABC algorithm is taken as the standard of the other algorithms, and the same random seed is run for initialization. All algorithms have the same number of population and total fitness evaluation. Each algorithm is repeated for three times. These three runs use different random seeds.

QPSO, DE, ABC and GDABC are initialized by (1) and have the same random seeds, but due to the complexity of the

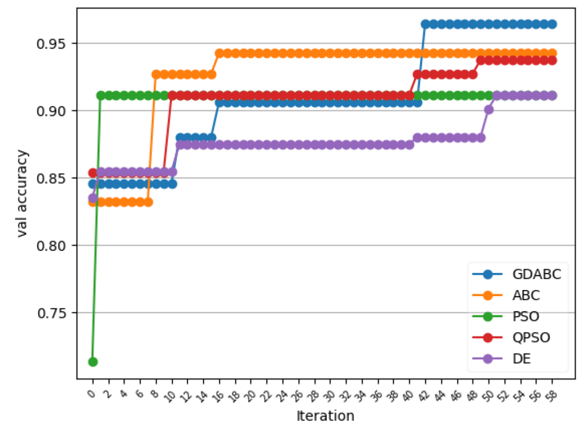

Fig. 5. The val accuracy of each SI algorithm along iteration on the MDVR-KCL dataset for the first run. The number of iterations is 59.

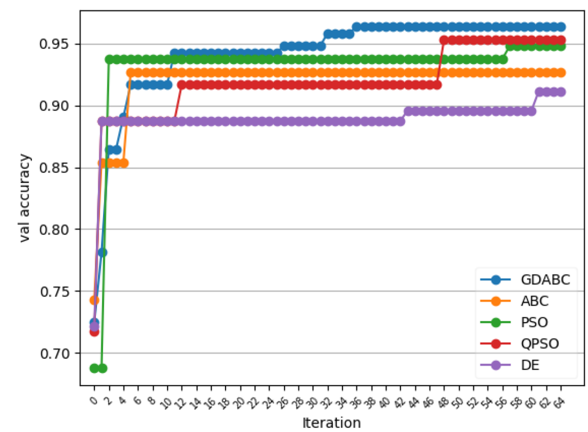

Fig. 6. The val accuracy of each SI algorithm along iteration on the MDVR-KCL dataset for the second run. The number of iterations is 65.

model and the parallelism of training, the initial fitness is not exactly the same. PSO adopts the different initialization method, $x^{i,j} = uniform(x_j^{\min}, x_j^{\max})$, so the initial values are different. It can be seen from Figs. 5, 6, and 7 that GDABC does not perform well in the early stage. Based on our proposed range prune strategy and the dimension adjustment strategy especially in the later stage, the algorithm achieves the highest accuracy in the verification set, which proves that our proposed strategies are effective. In particular, compared with ABC algorithm, GDABC algorithm has significantly improved exploring ability and the ability to jump out of local optima. At the same time, validation accuracy can reach more than 96% and finally remain stable in fewer iterations. These demonstrate the GDABC algorithm can converge in limited iterations. Table IV shows the best values of the 8 hyperparamers by GDABC algorithm at each run and Table V shows the corresponding accuracy of the models. As can be seen from Table IV, hyperparameter configuration obtained by each run is not exactly the same, which has a great relationship with the random initial value of the nectar. So when comparing with the other optimization algorithms, the same random seed is used to prevent the initial value from affecting the final result of the algorithm. From Table V, the training accuracy

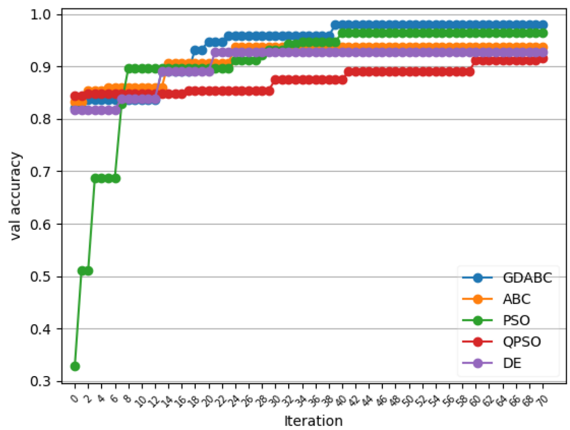

Fig. 7. The val accuracy of each SI algorithm along iteration on the MDVR-KCL dataset for the third run. The number of iterations is 71.

TABLE IV
THE BEST VALUES OF 8 HYPERPARAMETERS FOUND BY GDABC

Run	Best Hyperparameter							
	x_0	x_1	x_2	x_3	x_4	x_5	x_6	x_7
1	5	SGD	tanh	average	0.005	0.918	0.007	0.352
2	7	SGD	tanh	average	0.004	0.937	0.005	0.405
3	5	SGD	tanh	max	0.005	0.930	0.009	0.354

TABLE V
THE ACCURACY OF THE MODELS UNDER HYPERPARAMETRIC CONFIGURATION

Run	train acc	val acc	test acc
1	100	96.4	95.2
2	100	96.3	97.4
3	100	97.5	98.0

of ResNet50 models optimized by GDABC algorithm is 100% and their validation accuracy and test accuracy can reach 96% on average. These mean that for non-experts, it is easier to optimize hyperparameters of ResNet50 with the GDABC algorithm to get a model that performs well on the MDVR-KCL dataset than manual adjustment. It can be seen from the above that our proposed GDABC optimization algorithm is effective.

Experiment 2. Compared with current Parkinson's diagnosis works on the MDVR-KCL dataset. At present, the diagnosis of Parkinson's mainly uses MFCC to extract sound features, and uses traditional classifiers such as SVM and KNN for classification. In recent years, scholars have found that Vggish performs well in sound feature extraction [52], so we also include it in the comparison. All methods compared with our work have the same network parameter settings.

Table VI shows the comparison results between the current works and ours. Under the same preprocessing, it is obvious that the network model optimized by GDABC shows better generalization ability compared with ResNet50 without hyperparameter adjustment. In addition, even if Vggish is used for feature extraction, both SVM and KNN as Parkinson's classifiers

TABLE VI
COMPARISON RESULTS WITH CURRENT WORKS ON THE MDVR-KCL DATASET

Method	Train Acc	Val Acc	Test Acc
Vggish+SVM	66.8	-	60.5
Vggish+KNN	**100**	-	46.7
ResNet50	98.8	89.1	87.0
Kurada [50]	-	-	87.0
Huang [31]	97.3	-	-
Toye [51]	-	-	90.7
ResNet50+GDABC	**100**	**96.4**	**96.6**

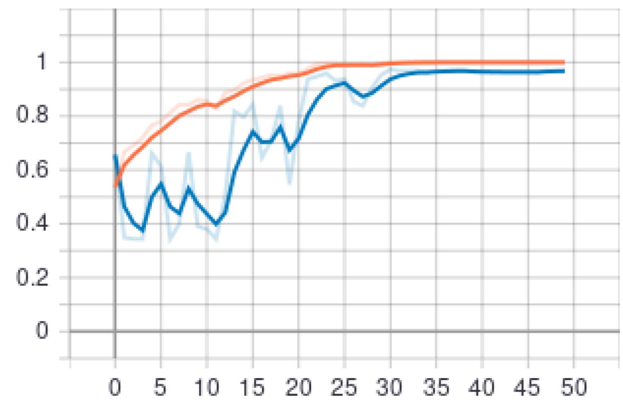

Fig. 8. The accuracy curve in the training of the best network model found by GDABC algorithm of the third run.

have not achieved ideal classification results. KNN classifier has high training accuracy, but poor generalization ability. It can be seen that the interpretability of the model is not strong, and the data imbalance leads to misclassification. For SVM classifier, the accuracy of training and testing are both low, which may be due to its high requirements for parameter and data. Huang et al. [31] used parseimouth library and Praat, two Python libraries to extract features of the MDVR-KCL dataset, and used decision tree and ResNet50 to classify Parkinson disease. ResNet50 got better results, and the training accuracy was 97.3%. Kurada et al. [50] pre-trained the Vggish model in millions of audio clips from the YouTube sound dataset, extracted the features of the dataset based on Mel spectrum and used KNN as the classifier. The final test accuracy on the MDVR-KCL dataset was 87%. Toye and his collaborators [51] also used MFCC to process the MDVR-KCL dataset, and compared seven classification models such as k-nearest neighbor, decision tree and support vector machine. The results demonstrated that the classification accuracy of decision tree was the highest, up to 90.7%. The results demonstrate that our method has higher classification accuracy and better robustness, proving the effectiveness of our entire Parkinson's auxiliary diagnosis system.

After each running, our proposed auxiliary diagnosis system will finally retrain the model and evaluate the accuracy of the model on the test set. Fig. 8 illustrates the result of performance testing for GDABC algorithm in the third run. Red curve indicates the accuracy of training and blue one indicates

TABLE VII
THE COMPARISON RESULTS OF SENSITIVITY, SPECIFICITY, PRECISION AND F1-SCORE ON THE MDVR-KCL DATASET

Method	Sen	Spec	Pre	F1
Vggish+SVM	7.1	96.0	54.5	12.6
Vggish+KNN	38.1	52.4	34.8	36.4
ResNet50	89.5	89.6	83.0	86.1
Huang [31]	72.3	-	-	-
Toye [51]	**95.2**	85.2	92.8	88.7
ResNet50+PSO	91.3	95.5	91.3	91.3
ResNet50+QPSO	92.7	88.8	81.0	86.5
ResNet50+DE	81.1	94.0	87.5	84.2
ResNet50+ABC	88.4	**96.3**	92.4	90.3
ResNet50+GDABC	94.2	95.5	**93.1**	**93.6**

the accuracy of verification. The final verification accuracy is almost the same as the training accuracy which tends to 1, showing the performance superiority of the network after adjusting parameters. To further prove the effectiveness of our algorithm, we not only use the fitness function accuracy of the algorithm as an indicator, but also use several other indicators to evaluate all the experimental results, such as as Sensitivity, Specificity, Precision and F1-Score. Their calculation formulas are presented as Formula 7 - 10. Among them, True Positive means that the prediction is positive and the prediction is correct, that is, its true label is positive. False Negative means that the prediction is negative, but the prediction is incorrect, that is, its true label is positive. Similarly, False Positive means the prediction is positive but the label is negative, and True Negative means the prediction is negative and the label is negative. In this paper, positive and negative represents PD and healthy respectively. The comparison results of these indicators are shown in Table VII. Using Vggish to extract features, the overall indicators of KNN classifier are not high, which may be caused by model instability. The effect of SVM classifier is poor, and almost all samples are predicted to be negative, which may be due to the wrong selection of parameters and high requirements for data. Due to its fast convergence, the overall indicators of PSO are slightly better than those of standard ABC algorithm. The overall indicators of DE and QPSO algorithm are slightly worse, basically in line with the standard ResNet50. The proposed GDABC algorithm is not only better than the standard ResNet50, but also further improved than the standard ABC algorithm in general. However, because the convergence speed of GDABC algorithm is not fast enough, the optimized model is slightly inferior to the method [51] in sensitivity. In summary, it can be seen from Table VII that our method is superior to the other compared methods in terms of these indicators on the whole, which proves the effectiveness and superiority of our auxiliary diagnosis system.

1) Sensitivity
Sensitivity measures the ability of the classifier to identify positive examples. In this paper, it represents the proportion of correctly predicted Parkinson patients in all true Parkinson patients. Its value is the same as the recall rate.

$$Sen = \frac{True\,Positive}{True\,Positive + False\,Negative} \tag{7}$$

2) Specificity
Specificity measures the ability of the classifier to identify negative examples. In this paper, it represents the proportion of those diagnosed as healthy among all healthy volunteers.

$$Spe = \frac{True\,Negative}{True\,Negative + False\,Positive} \tag{8}$$

3) Precision
Precision is the proportion of correctly predicted Parkinson patients in all volunteers predicted to be Parkinson patients.

$$Pre = \frac{True\,Positive}{True\,Positive + False\,Positive} \tag{9}$$

4) F1-Score
Sometimes there are contradictions between Precision and Recall indicators, and F1-Score is to comprehensively evaluate the precision rate and recall rate.

$$F1 = \frac{2 \times Precsion \times\ Recall}{Precsion + Recall} \tag{10}$$

V. CONCLUSION

To find Parkinson patients and treat them as soon as possible, we propose a Parkinson's auxiliary sound diagnosis system which uses the proposed GDABC algorithm to optimize the hyperparameters of ResNet50. The optimized network performs feature extraction on the spectrogram and acts as a classifier to achieve Parkinson's classification on the MDVR-KCL dataset. Proposed GDABC algorithm adds "Mixed coding strategy," "Range Prune strategy" and Dimension adjustment strategy" into basic ABC algorithm. "Mixed coding strategy" maps different types of hyperparameters to continuous domain. "Range pruning strategy" is to narrow scope of search and "Dimension adjustment strategy" is to adjust gbest dimension by dimension. For integer hyperparameters, update with the help of $Pruned\,Range$. For floating-point ones, use the current best value as the mean, and use $Gbest\,Distance$ as the variance to implement Gaussian randomization to update. GDABC algorithm improves the local search ability and reduces time complexity of basic ABC algorithm. Compared with the other algorithms, our algorithm shows better performance in optimizing hyperparameters of ResNet50. It also illustrates that classification accuracy of our work is better than some current works. Our work shows that this method has strong feature extraction ability and can improve the accuracy of diagnosis. At present, the time consuming of the training of deep network based on ABC algorithm is expensive. In the future, we will develop a lightweight hyperparameter optimization method to shorten the time consuming and and bring the auxiliary diagnosis system to clinical diagnosis in real life.

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